



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO COCO SALEM-PCM PETROLEUM AGENCIES U

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

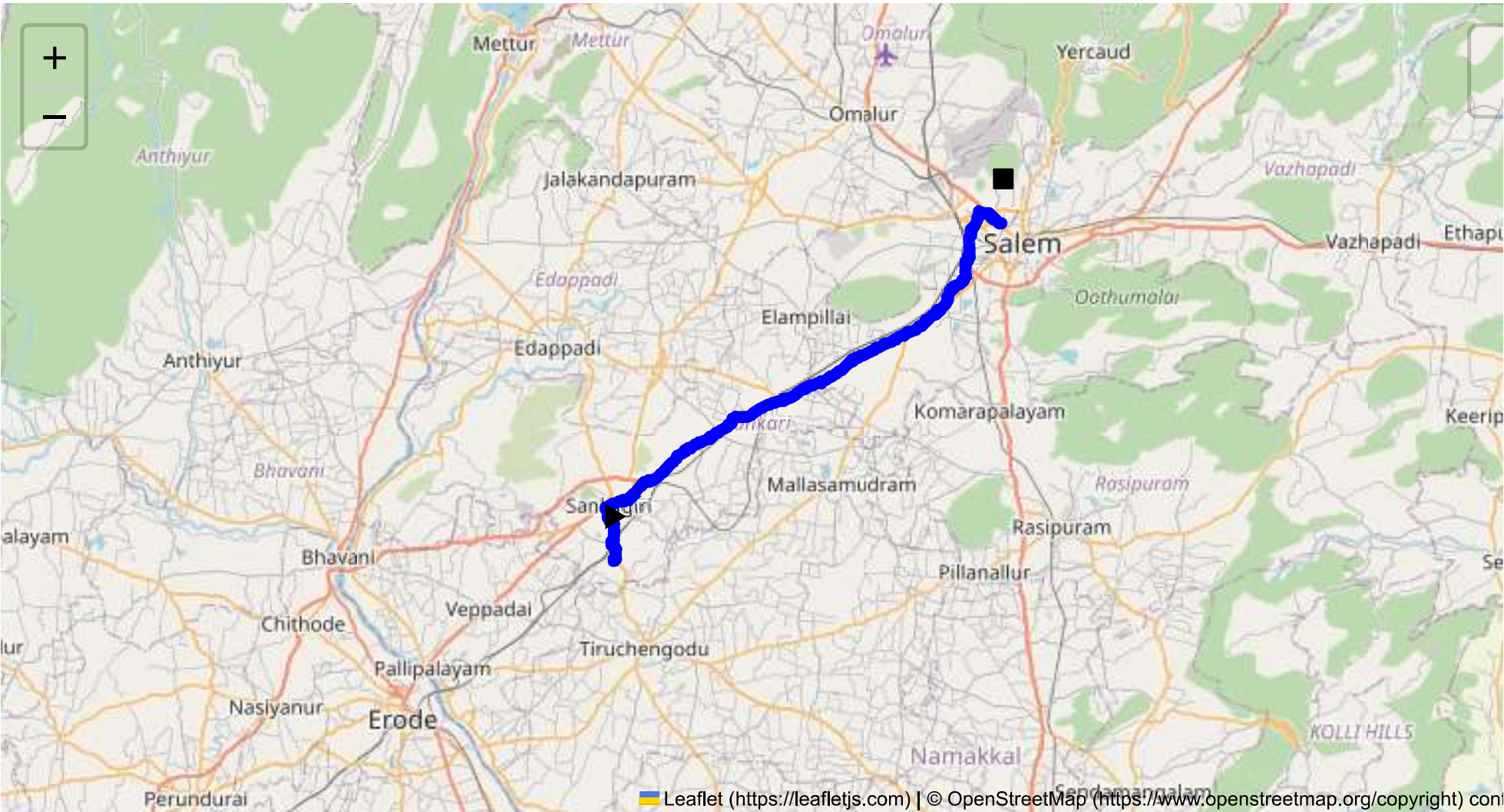
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 44.99 km
Estimated Duration: 1.0 hours
Adjusted Duration (Heavy Vehicle): 1.2 hours
Start: (11.4381, 77.8734)
End: (11.667741, 78.1429683)



Welcome to the Journey Risk Management Study

To analyze the safety of the route from Sangagiri, Tamil Nadu 637301, to Salem, Tamil Nadu 636002, for truck drivers transporting hazardous materials, I have compiled the following comprehensive overview:

1. Overview of the Route Map

The route primarily follows NH 544 (part of the Salem-Kochi highway), transitioning onto local roads as it approaches the destination in Salem. This is a heavily used commercial route due to its proximity to Salem, a significant city in Tamil Nadu.

2. Typical Weather Conditions and Potential Weather-Related Hazards

- The area experiences a tropical climate with three main seasons: hot, wet, and cool.
- During the monsoon season (June to September), heavy rains can cause waterlogging and poor visibility.
- Summer months can be extremely hot, affecting vehicle performance and driver alertness.
- Fog may occur during early morning hours in winter months.

3. Traffic Patterns, Highlighting Peak Hours and Congestion-Prone Areas

- Peak traffic hours are typically 8-10 AM and 5-8 PM.
- Congestion is common at the junctions near Sankari and while entering Salem city.
- Weekends might experience increased traffic due to local travel.

4. Assessment of Road Quality and Infrastructure

- NH 544 is generally well-maintained with adequate signage. However, transitions to local roads can feature narrow lanes and less reliable maintenance.
- Some sections might have potholes, especially after the monsoon season.

5. Suggestions for Alternative Routes for Emergencies

- An alternative is to take the SH 86 through Sankari, which runs parallel to NH 544 for some distance.
- Another is moving via Omalur to bypass some congestion areas before entering Salem from a different direction.

6. Summary of Local Regulations Affecting Hazardous Material Transport

- Transport of hazardous materials requires adherence to regulations on permissible timing and truck safety standards.
- Restrictions apply on carrying certain types of hazardous materials through populated areas, which may necessitate detours.

7. Overview of Historical Incidents Involving Heavy Vehicles or Hazardous Materials

- Minor incidents have been reported involving heavy vehicles, often due to driver fatigue or technical failures.
- There is no significant record of hazardous material accidents, though transporters should remain vigilant.

8. Environmental Considerations and Sensitive Areas

- Proximity to agricultural fields requires caution to prevent spills that could affect land or waterways.
- There are occasional wildlife crossings near less urbanized sections, requiring extra attention.

9. Analysis of Communication Coverage, Noting Potential Dead Zones

- The route generally has good communication coverage with sporadic dead zones between Sankari and Salem.
- Mobile coverage improves significantly near urban centers.

10. Estimated Emergency Response Times for Different Route Segments

- Urban areas near Salem offer quicker response times (20-30 minutes), while rural segments near Sangagiri might experience delays (30-60 minutes).
- It is advisable to keep an emergency response contact list updated and available at all times.

12. Overall Summary of Risk Assessment

The route is moderately safe for transporting hazardous materials, with no major safety obstacles that cannot be mitigated with careful planning and adherence to regulations. Weather and road conditions should be continuously monitored, especially during peak hours and adverse weather seasons. Emergency preparedness and communication plans should be solidified, ensuring quick response capability along all segments of the route.

Overall, with proper precautions and adherence to safety protocols, the route from Sangagiri to Salem remains manageable for heavy vehicle transportation.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	High	11.43811, 77.87348	15 KM/Hr
1	Turn	Medium	11.43956, 77.87340	30 KM/Hr
2	Turn	High	11.43968, 77.87345	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
3	Turn	High	11.44029, 77.87544	15 KM/Hr
4	Turn	High	11.45004, 77.87206	15 KM/Hr
5	Turn	Medium	11.45028, 77.87202	30 KM/Hr
6	Turn	High	11.47403, 77.86841	15 KM/Hr
7	Turn	Medium	11.47410, 77.86845	30 KM/Hr
8	Turn	High	11.49149, 77.89601	15 KM/Hr
9	Turn	High	11.49224, 77.89601	15 KM/Hr
10	Turn	Medium	11.60942, 78.10056	30 KM/Hr
11	Turn	Medium	11.61736, 78.10588	30 KM/Hr
12	Turn	High	11.64007, 78.11794	15 KM/Hr
13	Turn	Medium	11.64011, 78.11789	30 KM/Hr
14	Turn	High	11.67074, 78.12668	15 KM/Hr
15	Turn	High	11.67081, 78.12661	15 KM/Hr
16	Turn	High	11.67582, 78.12786	15 KM/Hr
17	Turn	Medium	11.67542, 78.13418	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	police	Police Quarters	11.4741243, 77.8673495	30 km/h	Medium
1	hospital	Kathir Eye Hospital	11.4740976, 77.8681591	30 km/h	Medium
2	hospital	Thilagam Hospital	11.4753573, 77.8685829	30 km/h	Medium
3	clinic	Meyyazhagan Clinic	11.4752254, 77.8685671	30 km/h	Medium
4	hospital	Dr. Jaganathan Hospital	11.4752717, 77.8709312	30 km/h	Medium
6	hospital	Sankari Hospital	11.4777496, 77.873892	30 km/h	Medium
8	clinic	Dr. Dhanasekar Clinic	11.4793113, 77.8829247	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
9	hospital	Sri Vellaiyan Hospital	11.4798876, 77.8831394	30 km/h	Medium
11	hospital	C. N. Hospital	11.55724, 78.0096	30 km/h	Medium
12	hospital	Annapoorna Medical College Hospital	11.57693, 78.037627	30 km/h	Medium
13	hospital	Vinayaka Mission's Homeopathic Medical College & Hospital	11.583617, 78.059145	30 km/h	Medium
14	hospital	Vinayaka mission Kirubananadah variyar medical college hospital	11.584984, 78.0618417	30 km/h	Medium
15	hospital	vims	11.5854239, 78.0639188	30 km/h	Medium
18	hospital	Vinayaka Mission’s Kirupananda Variyar Medical College	11.5977669, 78.0850992	30 km/h	Medium
22	hospital	Bharathi Hospital, Salem	11.6492647, 78.1204131	30 km/h	Medium
23	hospital	Sisu Hospital Inida Ltd	11.6505276, 78.1205003	30 km/h	Medium
24	hospital	Aishwarya Hospital	11.6716291, 78.1267807	30 km/h	Medium
25	hospital	C.M. Hospital	11.6747345, 78.1255034	30 km/h	Medium
26	pharmacy	Healthy Pharmacy	11.6762714, 78.127744	30 km/h	Medium
27	pharmacy	Thulasi Pharmacy	11.6761991, 78.1277377	30 km/h	Medium
29	hospital	Venkatesh Hospital and Urology Centre	11.6759517, 78.1324029	30 km/h	Medium
30	hospital	R.K. Hospital, Salem	11.6746905, 78.1339717	30 km/h	Medium
31	clinic	Bismillah Piles Clinic	11.6755285, 78.1343325	30 km/h	Medium
32	hospital	Kurinji Hospital, Salem	11.6771806, 78.1351224	30 km/h	Medium
33	hospital	Bhavani Hospital	11.6745697, 78.1362683	30 km/h	Medium
34	clinic	Ayush Care Siddha Centre	11.6731195, 78.1351967	30 km/h	Medium
35	hospital	Sri Gokulam Hospital Pvt Ltd	11.6725694, 78.1337169	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
36	clinic	Arunagiri Ent Nursing Home	11.6762823, 78.1369593	30 km/h	Medium
37	clinic	Srimaya Child Health Centre	11.6750852, 78.1370646	30 km/h	Medium
38	hospital	Gl Hospital	11.6742597, 78.1379092	30 km/h	Medium
39	clinic	Dr. Kanaks Skin Hair & Cosmetology Clinic	11.6722777, 78.1375557	30 km/h	Medium
40	hospital	Balaji Skin Hospital	11.676166, 78.1379488	30 km/h	Medium
41	hospital	Rainbow Hospital	11.6722366, 78.13774	30 km/h	Medium
42	hospital	Dr. Sasi Eye Care	11.6743698, 78.1379234	30 km/h	Medium
43	hospital	Dr Agarwal Vasns Eye Hospital, Salem	11.6700596, 78.1381854	30 km/h	Medium
44	police	Palapatti Police Station	11.669908, 78.1395206	30 km/h	Medium
45	police	Palapatti B 10 Police Station	11.6699556, 78.1394849	30 km/h	Medium
46	hospital	S S Medical Centre	11.6706585, 78.1398415	30 km/h	Medium
47	hospital	Saravana E.N.T Hospital	11.6720221, 78.1394629	30 km/h	Medium
48	hospital	Dr Sudhakar Ent Care Centre	11.6713519, 78.1399383	30 km/h	Medium
49	hospital	Kattabomma Hospital	11.6689589, 78.1371173	30 km/h	Medium
50	hospital	Ark Hospital	11.6716723, 78.1403218	30 km/h	Medium
51	hospital	Vasan Eye Care Hospital, Salem	11.6702294, 78.1407532	30 km/h	Medium
52	hospital	Dr. Agarwal Vasans Eye Hospital	11.6696681, 78.1417073	30 km/h	Medium
53	hospital	Pathanjali Ayurvedha Hospital	11.671244, 78.141525	30 km/h	Medium
54	hospital	S K S Hospital India Pvt Ltd	11.6701839, 78.1433663	30 km/h	Medium
55	hospital	Lotus Eye Care Hospital, Salem	11.6689406, 78.1425789	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
56	hospital	Pranav Hospitals Pvt Ltd	11.6693372, 78.142532	30 km/h	Medium
57	hospital	Sudha Hospital	11.6691694, 78.142588	30 km/h	Medium
58	hospital	Dr. S. Rajendran Children Hospital	11.6707079, 78.1421878	30 km/h	Medium
59	hospital	Eka Hospital	11.6691926, 78.1425812	30 km/h	Medium
60	clinic	Homeopathy Clinic	11.6691556, 78.1434845	30 km/h	Medium
61	clinic	K.S. Siddha & Varma Clinic	11.6674481, 78.1392422	30 km/h	Medium
62	hospital	S.K.S. Hospital	11.670548, 78.143109	30 km/h	Medium
63	clinic	Salem Child Care Centre	11.6706006, 78.1421572	30 km/h	Medium
64	hospital	RJR Hospital	11.6719465, 78.1432779	30 km/h	Medium
65	clinic	JH Cardiac Centre	11.6707391, 78.1421402	30 km/h	Medium
66	clinic	Salem Poly Clinic	11.6663605, 78.144658	30 km/h	Medium
67	hospital	KS Siddha & Varma Hospital	11.666952, 78.141606	30 km/h	Medium
68	hospital	S.P.S. Hospital	11.6669469, 78.1445952	30 km/h	Medium
69	hospital	Sri Vaari Raja Sidha Hospital	11.6668535, 78.1422245	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
5	school	Goverment Boys Higher Secondary School	11.477659, 77.8659632	30 km/h	Medium
7	marketplace	Sunday Market	11.4796502, 77.8708458	30 km/h	Medium